System_Architecture_Overview

Architecture Document

保险公司 技术开发部

Contents

System Architecture Overview

1. Document Objective

This document defines the overall architectural view and design principles for the insurance company's technology landscape. It serves as a unified framework for all subsystems, technology decisions, and architecture governance. The audience includes architects, tech leads, and developers.

2. Architecture Design Principles

Principle	Description	Reference
High Availability	Core system availability ≥ 99.99%, annual downtime < 53 min	Ant Cloud Standard
Disaster Recovery	Active-active across ≥ 3 data centers in ≥ 2 regions	Industry Best Practice
Distributed Microservices	Domain-oriented decomposition, independent deployment & scaling	Spring Cloud / K8s
Loose Coupling	Async messaging + standard APIs between domains, no direct DB sharing	DDD + CQRS
Observability	Full-trace Trace, Metrics, Logging pillars	OpenTelemetry
Security & Compliance	MLPS 2.0 / Personal Information Protection Law / C-ROSS	Regulatory requirements

3. Architecture Layers

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┌──────────────────────────────────────────────┐
│ Channel Access Layer │
│ App · Web · Agent · Broker · Bank · 3rdP │
└───────────────────────┬──────────────────────┘
│ API Gateway / WAF / CDN
┌───────────────────────┴──────────────────────┐
│ Business Middleware │
│ ┌─────┐ ┌─────┐ ┌─────┐ ┌─────┐ ┌─────┐ │
│ │Product│Underw│Risk │Claims│Renew│ │
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│ Technology Middleware │
│ Registry · Config · Gateway · MQ · Scheduler│
│ Logging · Monitoring · Tracing · CI/CD │
└───────────────────────┬──────────────────────┘
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┌───────────────────────┴──────────────────────┐
│ Data Platform │
│ OLTP → Canal → Kafka → Flink → ClickHouse │
│ Data Lake · DWH · Metrics · AI/ML │
└───────────────────────┬──────────────────────┘
│
┌───────────────────────┴──────────────────────┐
│ Infrastructure │
│ K8s · Storage · Network · Security · CDN │
└──────────────────────────────────────────────┘
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4. Core System Inventory

System	Responsibilities	Key Capabilities
Product Center	Product definition, rate factors, clause management	Rule engine, pricing model
Underwriting Center	Application entry, risk assessment, policy issuance	UW rule engine, image OCR
Policy Servicing	Endorsement, reinstatement, cancellation	Service rules, fund calculation
Claims Center	FNOL, investigation, assessment, settlement	Risk model, anti-fraud engine
Renewal Center	Renewal reminders, grace period management	Payment interface, SMS
Reinsurance Center	Facultative/treaty, ceded/assumed	Contract management, billing
Channel Center	Agent, broker, bancassurance, online	Commission rules, channel agreements
Payment Center	Premium collection, claim payment, reconciliation	Funding channels, error handling
Customer 360	Customer info, policy relationships, KYC	Unified customer view
Data Platform	DWH, reporting, regulatory, BI analytics	ETL, metric management

5. Key Business Process Flows

5.1 Underwriting Main Flow

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Sales → Application Entry → Auto/Manual UW → Payment → Policy Issue → Effective
```

- Cross-system dependency chain: Product → UW → Risk Engine/Image → Payment → Policy Issue
- Non-functional: Peak season TPS ≥ 2000, P99 < 500ms

5.2 Claims Main Flow

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FNOL → Investigation → Case Open → Assessment → Calculation → Approval → Settlement → Payment
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- Cross-system dependency: Claims → Risk Engine → Payment → Reinsurance (if exceed retention)
- Non-functional: Settlement completion ≤ 3 days (fast-track ≤ 1 day)

6. Architecture Governance

- **Architecture Review Board**: Monthly review of cross-domain designs and technology selections
- **ADR**: All decisions recorded with context, options, decision, and consequences
- **Tech Radar**: Quarterly update of technology adoption/retirement status
- **Compliance Check**: Architecture rule scanning integrated into CI pipeline

7. References

- Microservice Architecture Standard (this directory)
- Domain-Driven Design Standard (this directory)
- Data Architecture Standard (this directory)
- Security Architecture Standard (this directory)
- Technology Selection & ADR Standard (this directory)